

MAT1375, Classwork2, Fall2026

Ch2. Functions via Formulas

1. Let $f: A \rightarrow B$ be a function where A is the input / domain and B is the output / codomain. If the formula of f is given, then an input $a \in A$ can find an output $b \in B$ such that $b = f(a)$.
 $\downarrow$ belongs to $f(a) = b$

2. Given a function $f(x) = x^2 + 4x - 9$. Find the value of (a) $f(2)$; (b) $f(-3)$; (c) $f(0)$; (d) $f(h)$

$x=0 \leftarrow$ input replace "x" by "any numbers or notations"

$$f(0) = 0^2 + 4(0) - 9 = -9 \leftarrow \text{output}$$

$$(a) f(2) = (2)^2 + 4(2) - 9 = 4 + 8 - 9 = 3$$

$$(b) f(-3) = (-3)^2 + 4(-3) - 9 = 9 - 12 - 9 = -12$$

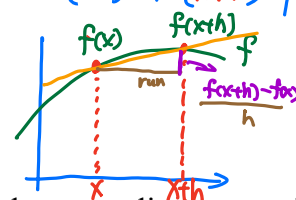
$$(c) f(h) = (h)^2 + 4(h) - 9 = h^2 + 4h - 9$$

$$f(x+h) = (x+h)^2 + 4(x+h) - 9$$

3. Difference Quotient

Let $y = f(x)$ be a function. We called the expressions

$$\frac{f(x+h)-f(x)}{h} \quad \text{or} \quad \frac{f(x)-f(a)}{x-a}$$



Difference Quotient for the function f (which represents the slope of the secant line connecting two points on a function's graph, or the average rate of change of the function over a small interval $[x, x+h]$).

4. Given a function $f(x) = x^2 + 4x - 9$. Find the value of (a) $f(x+h)$; (b) $f(x+h) - f(x)$; (c) $\frac{f(x+h)-f(x)}{h}$.

$$(a) f(x+h) = (x+h)^2 + 4(x+h) - 9$$

$$= (x+h)(x+h) + 4(x+h) - 9$$

$$= x^2 + xh + xh + h^2 + 4x + 4h - 9$$

$$= x^2 + 2xh + h^2 + 4x + 4h - 9$$

$$(c) \frac{f(x+h)-f(x)}{h}$$

$$= \frac{2xh + h^2 + 4h}{h}$$

$$= \cancel{h}(2x + h + 4) / \cancel{h}$$

$$= 2x + h + 4$$

$$(b) f(x+h) - f(x)$$

$$= (x^2 + 2xh + h^2 + 4x + 4h - 9) - (x^2 + 4x - 9)$$

$$= \cancel{x^2} + 2xh + h^2 + \cancel{4x} + 4h - \cancel{9} - \cancel{x^2} - \cancel{4x} + \cancel{9} = 2xh + h^2 + 4h$$

5. Piecewise(-Defined) Function

A piecewise function is a function whose domain is partitioned into several intervals on which the function may be defined differently.

$$|5| = 5$$

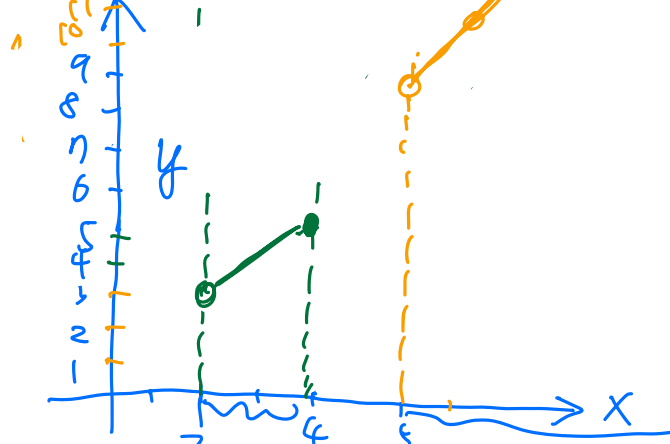
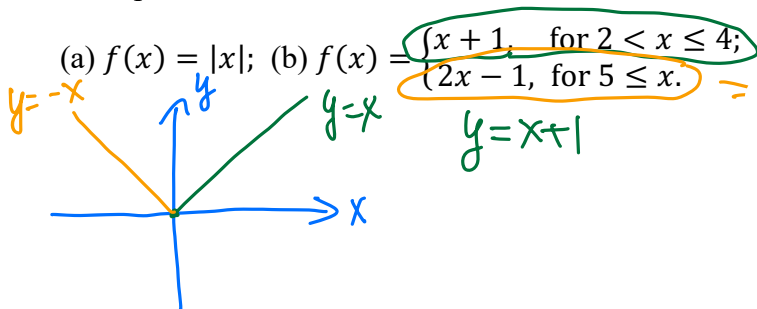
$$|-10| = 10$$

$$|2.5| = 2.5$$

$$f(x) = |x| = \begin{cases} x, & x \geq 0; \\ -x, & x < 0 \end{cases}$$

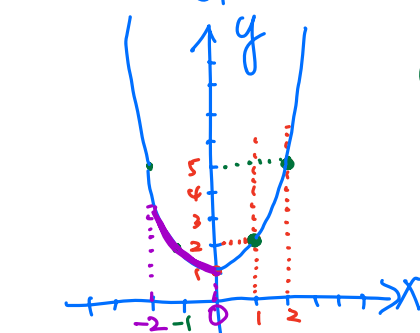
$$y = f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

6. Examples of Piecewise Functions



7. Consider the function described by the following formula. What is the domain of this function? Graph the function f .

$$f(x) = \begin{cases} x^2 + 1, & \text{for } -2 < x \leq 0; \\ x - 1, & \text{for } 0 < x \leq 2; \\ -x + 4, & \text{for } 2 < x \leq 5. \end{cases}$$

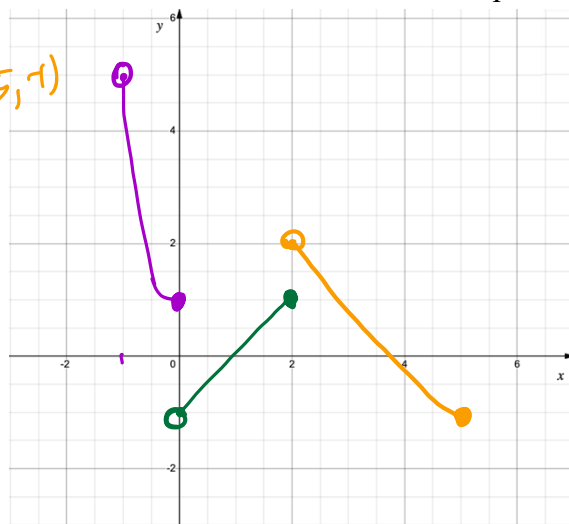


$$y = x - 1$$

$(2, 2)$ $(5, 1)$

x	2	5
y	2	1

$y = -x + 4$



$$\text{Domain} = \{x \mid -2 < x \leq 5\}$$

8. Find the domain of each of the following functions according to the standard convention of the domain.

(a) $f(x) = x^2 + 4x - 9$; (b) $f(x) = |x|$; (c) $f(x) = \sqrt{x}$; (d) $f(x) = \sqrt{x-3}$; (e) $f(x) = \frac{3x+5}{x-10}$.

Domain: check if any input gives nonsense output (s) $+\infty$, $-\infty$, complex numbers

(a) since all input in $\mathbb{R} = (-\infty, \infty)$, or all real numbers

have an output, the domain of $f(x) = x^2 + 4x - 9$ is all real numbers

$$D_f = \{x \mid x \in \mathbb{R}\}$$

(b) $D_f = \{x \mid x \in \mathbb{R}\}$ for $f(x) = |x|$

(c) $f(x) = \sqrt{x}$, $D_f = \{x \mid x \geq 0\}$

(d) $f(x) = \sqrt{x-3} \geq 0$

$$D_f = \{x \mid x \geq 3\} \quad \begin{aligned} x-3 &\geq 0 \\ \Rightarrow x &\geq 3 \end{aligned}$$

(e) $f(x) = \frac{3x+5}{x-10}$ $x-10 \neq 0$

$$D_f = \{x \mid x \neq 10, x \in \mathbb{R}\} \\ = \{x \in \mathbb{R} \mid x \neq 10\}$$